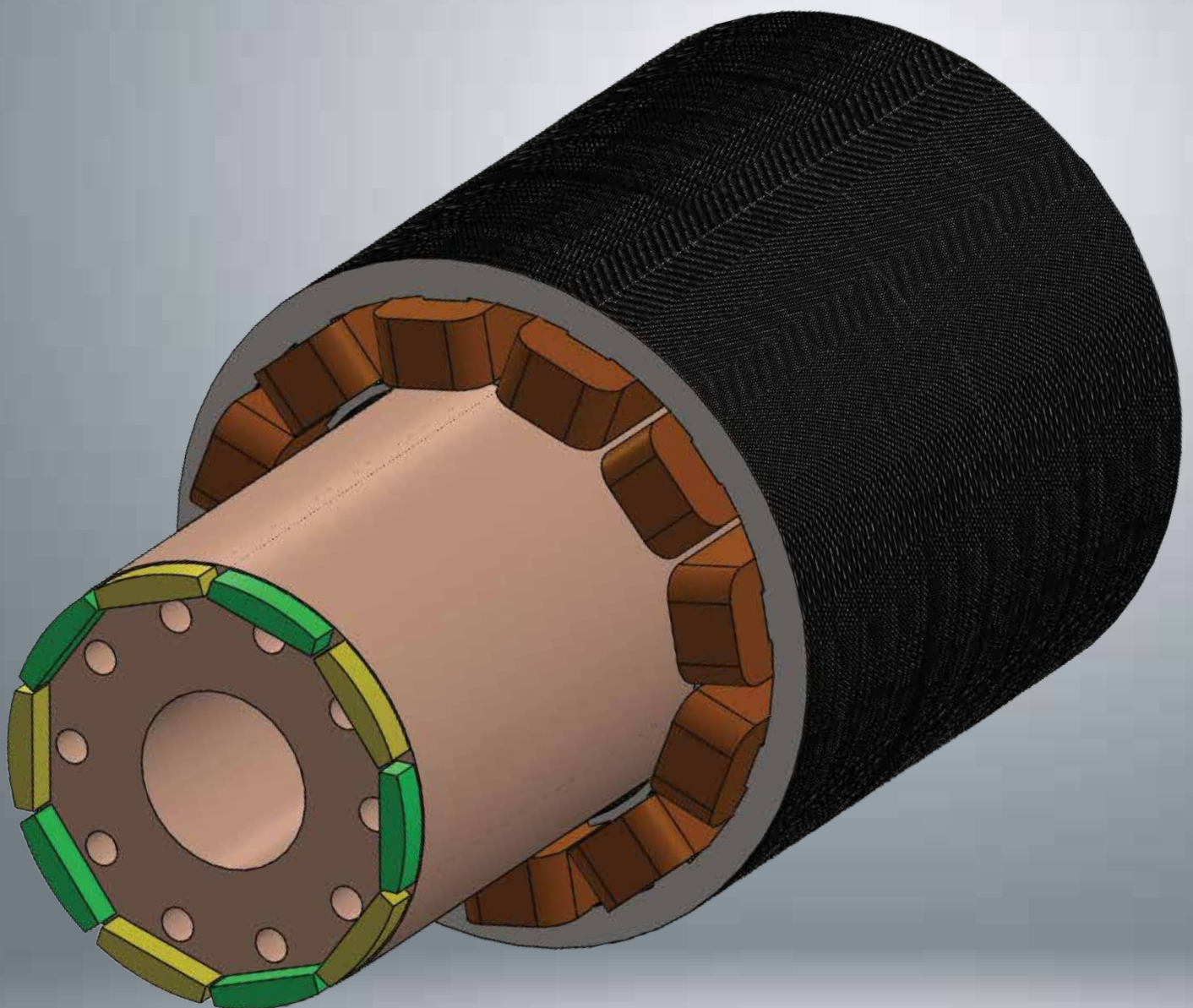




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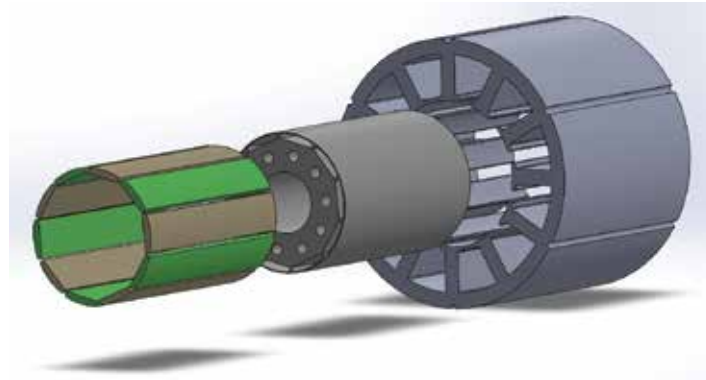
FRAMELESS BLDC MOTORS



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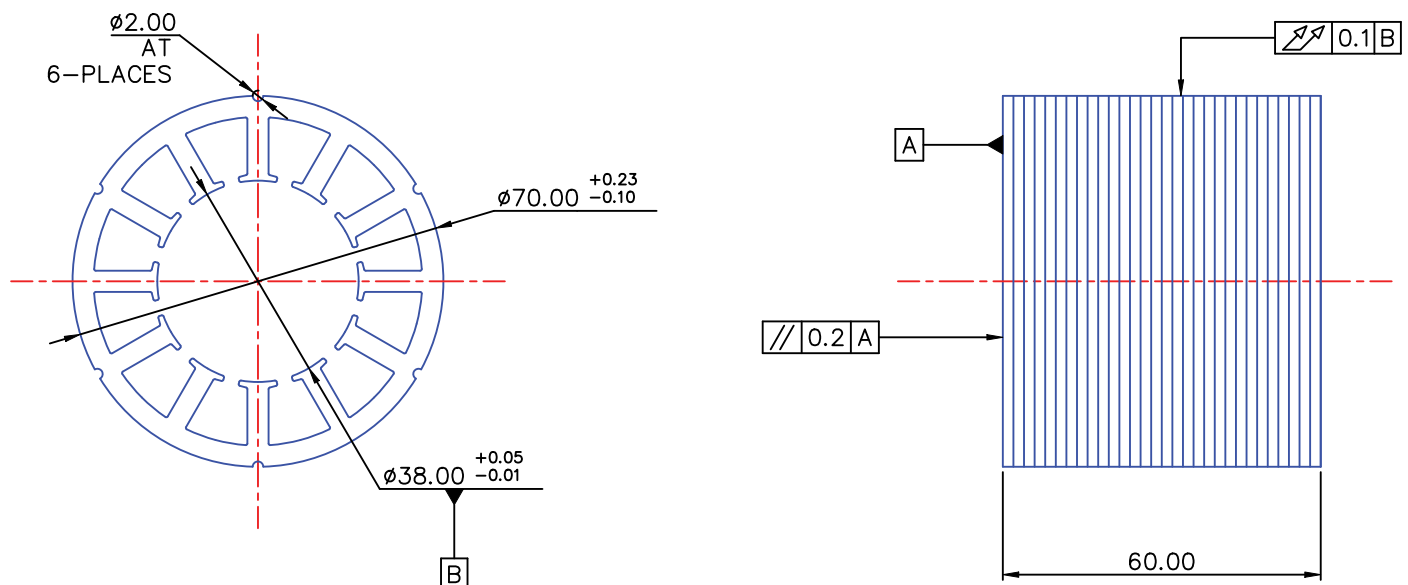
Overview

3Nm and 15Nm frameless BLDC motors are compact, high-performance torque motors designed for direct integration into custom mechanical assemblies. By eliminating the outer housing, shaft, and bearings, these motors offer high torque density, low inertia, and excellent efficiency.



MTBLDC3NI00V5800R	Design Parameters	
Supply Voltage – 100 Vdc	10 Pole 12 Slot	Core length – 60mm
Rated Torque – 3Nm	Stator OD – 70 mm	AMS S132 (nitriding capable steel)
Peak Torque – 4Nm (for 4 secs)	Magnet Material (N42UH)	Total Weight – 1.5kg
Rated Speed – >5800 rpm	Temperature Range – (-40to 120 °C)	Insulation Temperature – 150°C

3Nm FRAMELESS BLDC DUPLEX MOTOR SPECIFICATION – STATOR LAMINATION

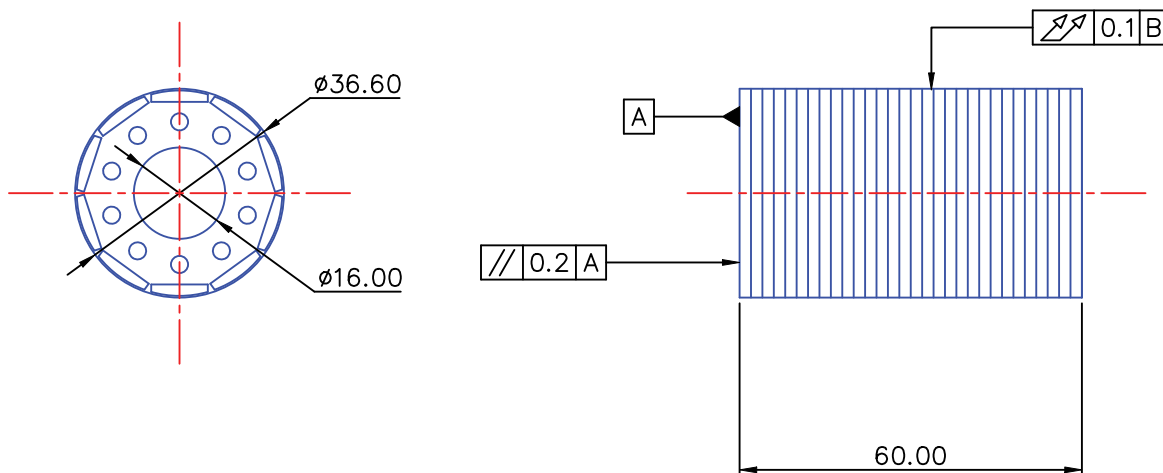


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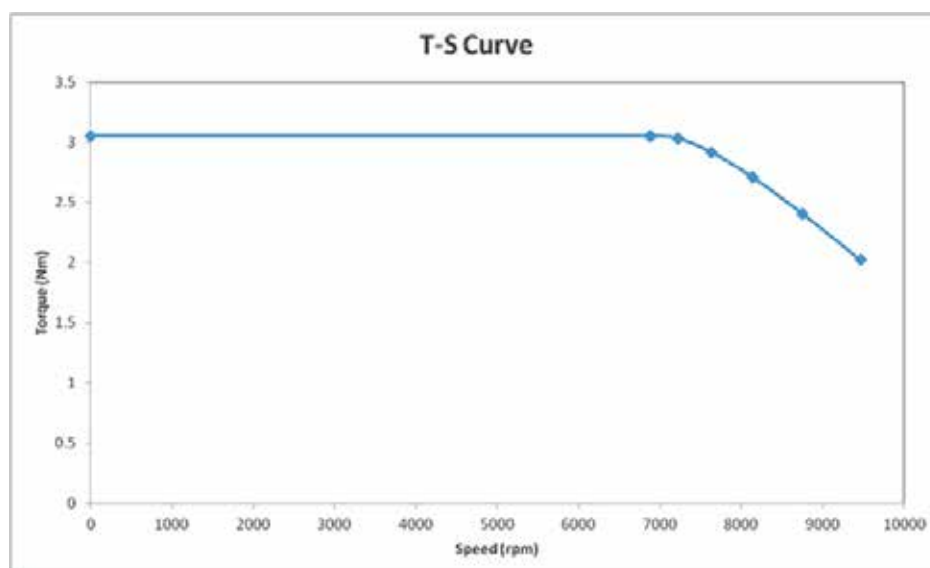
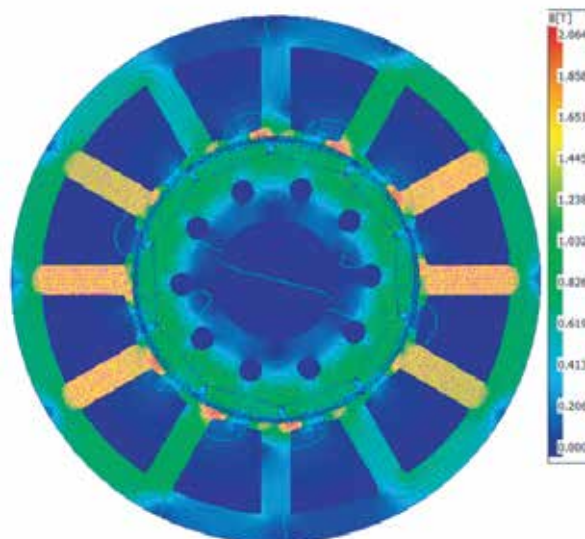
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3Nm FRAMELESS BLDC DUPLEX MOTOR SPECIFICATION – ROTOR LAMINATION



Full load flux density plot in Tesla



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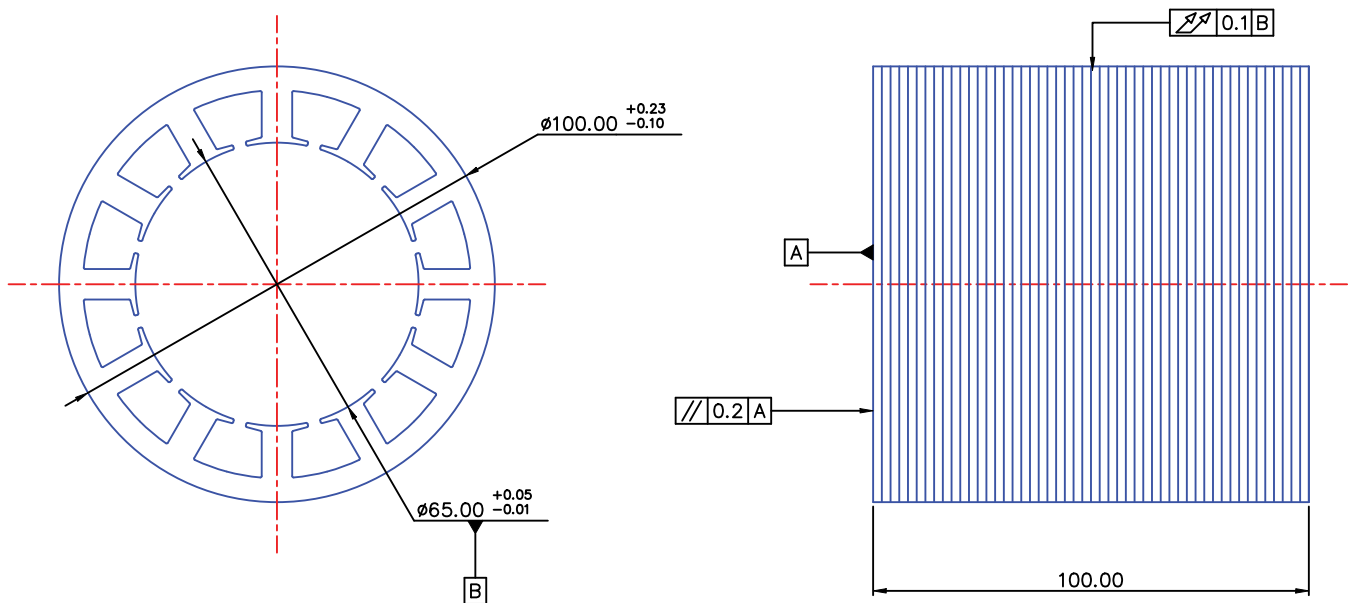
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Overview

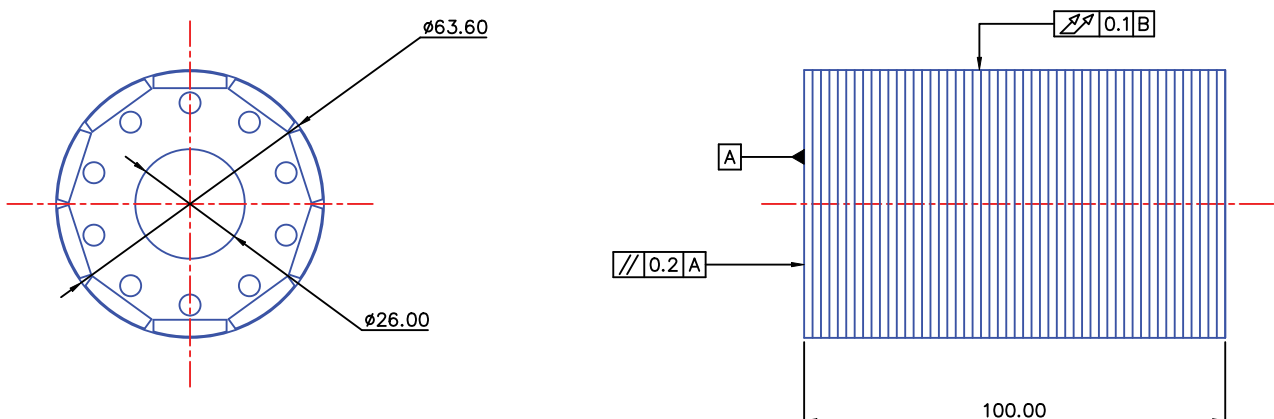
MTBLDC15N270V5200R
Supply Voltage – 270 Vdc
Rated Torque – 15Nm
Peak Torque – 20Nm (for 4 secs)
Rated Speed – >5200 rpm

Design Parameters	
10 Pole 12 Slot	AMS S132 (nitriding capable steel)
Stator OD – 100 mm	Magnet Material (N42UH)
Core length – 110mm	Total Weight – 5kg
Temperature Range – (-40to 120°C)	Insulation Temperature – 150°C

15Nm FRAMELESS BLDC DUPLEX MOTOR SPECIFICATION – STATOR LAMINATION



15Nm FRAMELESS BLDC DUPLEX MOTOR SPECIFICATION – ROTOR LAMINATION

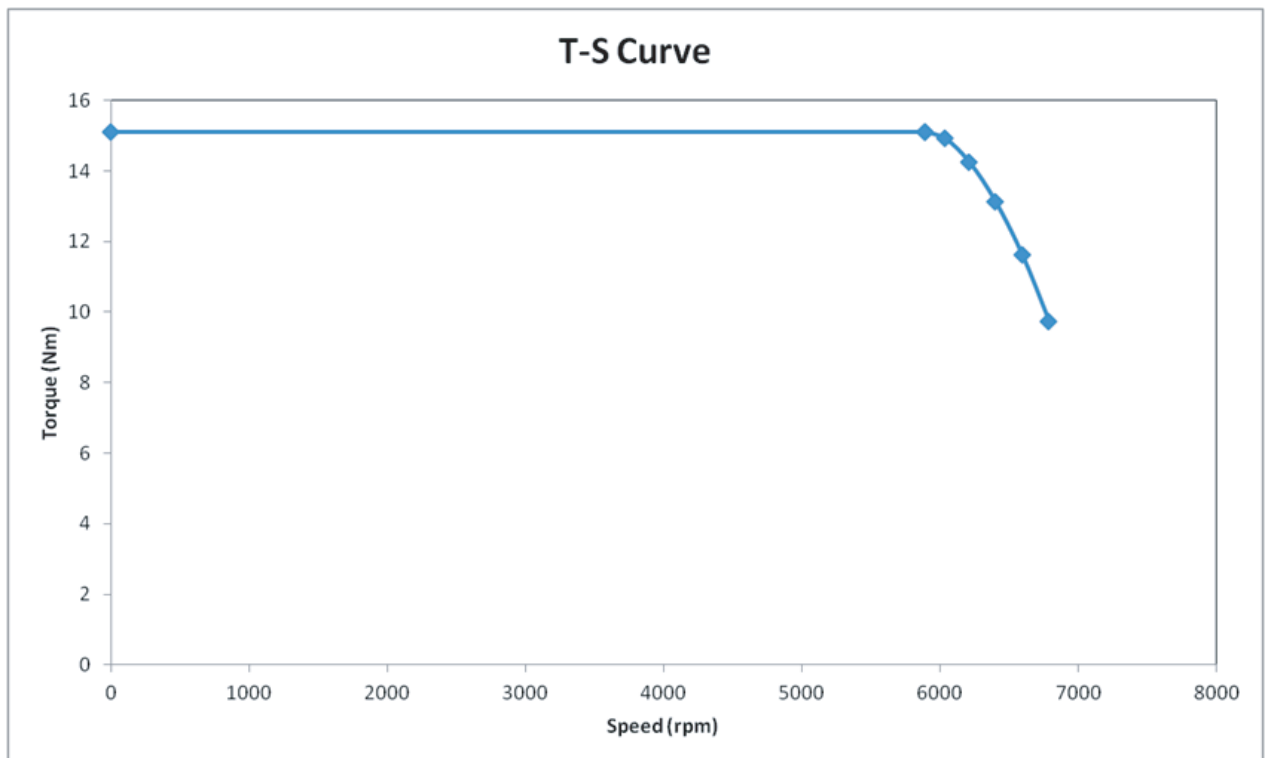
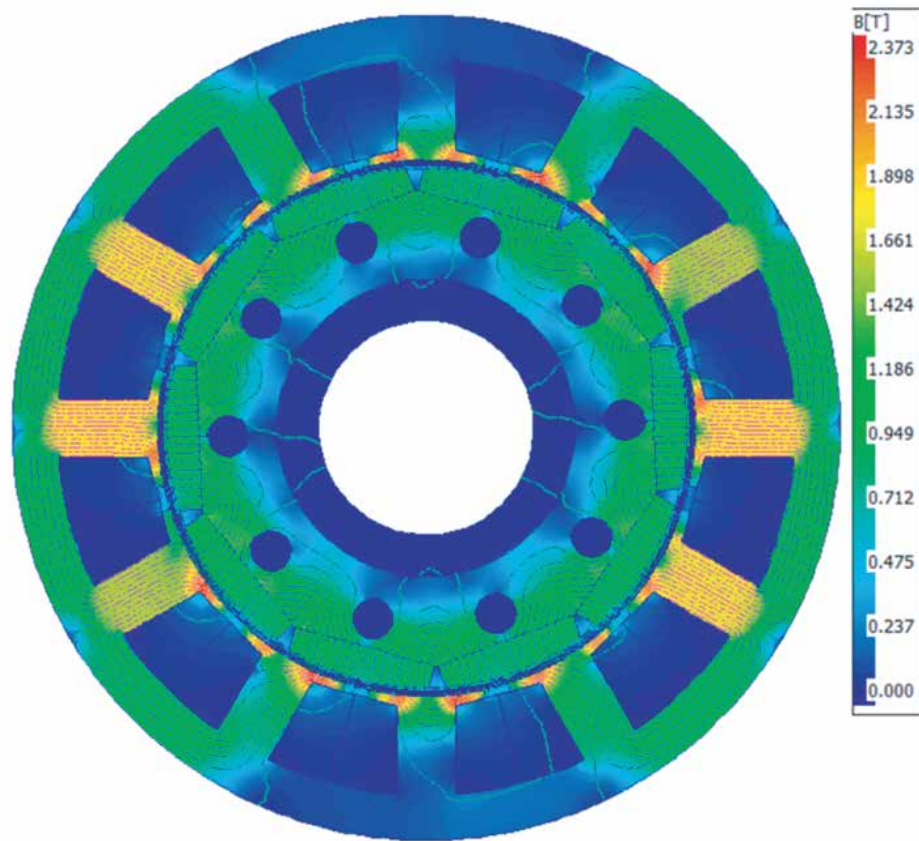


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Full load flux density plot in Tesla



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Standards & Material Grades :

Core Motor Kit (Stator–Rotor)			
Assembly	Part / Material	Aerospace/AMS Guidance	Key Specs / Notes
Stator laminations	Electrical steel lamination stack	Use low loss aerospace qualified electrical steel ; if cobalt alloy stamping is mandated, use AMS S132 (nitriding capable steel) for rotor/stator stampings per your spec; avoid silicon iron if prohibited	Thickness 0.20–0.35 mm; burr control $\leq 10\ \mu\text{m}$; grain orientation per design
Winding wire	Copper, enamelled	Source from aerospace approved supplier (e.g., SANGVI Aerospace per ADA). Insulation Class F/H . Conform to NEMA MW or equivalent aerospace spec; require material CofC and lot traceability	Gauge per current; enamel type: polyester imide/amide imide; thermal class $\geq 155\ ^\circ\text{C}$ (F) or $180\ ^\circ\text{C}$ (H)
Slot insulation	Polyimide/aramid (Nomex/Kapton)	AMS 36374/36371 equivalent polyimide film or aerospace qualified insulation	Dielectric $\geq 2\ \text{kV}$; thickness 0.10–0.30 mm
Lead wires	High temp wire (PTFE/silicone)	AS22759 series (PTFE/ETFE aerospace wire)	Voltage rating $\geq 600\ \text{V}$; color coded U V W
Impregnation resin	Epoxy/VPI resin	Aerospace qualified epoxy; Tg $\geq 120\text{--}150\ ^\circ\text{C}$; low volatiles; outgassing tested (NASA ASTM E595 if required)	VPI or trickle; thermal class F/H
Hall sensor PCB (optional)	PCB + 3 Hall ICs	IPC Class 3 PCB; components rated $-40\ ^\circ\text{C}$ to $+135\ ^\circ\text{C}$; conformal coat per IPC CC 830	120° electrical spacing; keyed connector
Temperature sensor	NTC/PT100/PT1000	Aerospace grade sensor; shielded leads per AS22759	Mount near end turns
Rotor back iron	Steel ring	AMS 2301/2303 cleanliness level on steel; anti corrosion coating	OD/ID per airgap
Permanent magnets	NdFeB arc segments	High reliability magnets with Ni/epoxy coating; demag margin verified; CofC + BH curve	Grade per torque; magnetization pattern per design
Magnet adhesive	Structural epoxy	Aerospace structural adhesive (e.g., EA class equivalent) with shear $\geq 20\ \text{MPa}$ and high temp cure; provide process spec & cure record	Bondline control shims/fixtures
Retention sleeve	Metallic or CFRP sleeve	Aerospace-alloy (e.g., AMS 5643 – 17 4PH) or CFRP with certified lay up; proof tested	Designed for burst margin at 5,000 RPM
Balancing weights	Fasteners/epoxy	Aerospace grade fasteners (e.g., MS/NAS series) or qualified epoxy	For fine balance
Stator mount features	Rings/lands	Housing interfaces per aerospace drawing control; fasteners NAS/AN series	Press fit or bolted
Rotor hub/adaptor	Hub/collet	AMS 5643 or AMS 7075 (if aluminum) with anodize; concentricity $\leq 0.02\ \text{mm}$ TIR	Coupling to user shaft
Phase harness	Cable & connector	Cable per AS22759 ; connector per MIL DTL series (as applicable)	Shielding if EMI risk
Signal harness	Cable & connector	As above; keyed/locking connector	Hall/Temp leads

Area	Part	Aerospace/AMS Guidance	Notes
Fasteners	Screws, washers	Use MS/NAS/AN fasteners per drawing; self locking per NASM 25027	Torque values & threadlocker specified
Shims/spacers	Precision shims	Aerospace qualified shim stock; traceable lot	Airgap setting
Thermal interface	Pads/grease	High k aerospace thermal materials (silicone pad) with CofC ; grease MIL SPEC equivalent	Stator–housing path
Insulation boots	Sleeves/boots	AMS 3656 PTFE or equivalent	Over crimp/termination
EMI/EMC	Shields/ferrites	Comply with DO 160 (if applicable)	Cable routing best practices
Feedback	Resolver/encoder	Aerospace resolver (MIL DTL 81963) or radiation tolerant encoder (as needed)	For high precision
Environmental protection	Potting/seal	Aerospace qualified potting; IP requirement defined	Moisture/particle ingress
Docs & data	ICD, specs	Drawings under config control (CM) ; mark CTQs	Revision discipline
Packaging	Protective	Anti mag shielding, ESD control; humidity indicator	Export controls awareness

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Standards & Material Grades :

Category	Requirement
Materials	Certificates of Conformance (CofC) for all AMS/ASTM materials; heat/lot traceability
Process	Documented adhesive bond and cure records; winding and impregnation travelers; torque logs
Electrical	Insulation resistance (IR), hi pot, phase resistance, back EMF constant, efficiency mapping
Balance	Rotor balancing to ISO 1940 G 1.0 (or program defined); records retained
Environmental	-40 °C to +135 °C functional tests (per your requirement) for brake, sensors, wiring; thermal cycling
EMI/EMC	If applicable to platform: pre compliance vs DO 160 sections for conducted/radiated emissions
Configuration mgmt	Unique serials; as built BOM; FAI (AS9102) for first article; PPAP like evidence if requested

Advanced Manufacturing Facilities for Motor Production

Automatic Coil Winding Machine  	Epoxy Potting Machine  	CNC VMC Machine  	CNC Turning Machine  	Laser Engraving & Marking Machine  
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Dyno**



**Dynamic Balancing &
Vibration Analyzer**



**Life Cycle / Endurance
Test Rig**



Contact

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{ Motors Manufacturing Plant }

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